FairPoC: Non-IID-Aware Proof of Contribution with Blockchain-Native Adversarial Defence for Auditable Federated Learning

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***Abstract*—Open federated learning (FL) marketplaces face two intertwined threats: unfair reward allocation that discourages high-quality participation, and rational adversaries who exploit contribution scoring to inflate their payoffs. Existing blockchain-FL proposals address at most one of these threats, and none provides a formally proven economic security guarantee. This paper presents FairPoC, a unified framework that resolves both threats through a single on-chain mechanism. The key insight is that the Proof of Contribution (PoC) score — a weighted combination of validation accuracy and median-clipped gradient norm — simultaneously serves as a convergence-stable fairness signal *and* tamper-proof slash evidence in the `FairPoC.sol` ERC-20 smart contract. Clients stake tokens before each round; the contract autonomously slashes any client whose PoC score falls below 30% of equal share and redistributes forfeited stakes to honest participants in the same atomic transaction, with no trusted intermediary. We make four contributions beyond our conference version [?]: (i) a formal Nash equilibrium proof showing that honest participation is the dominant strategy for any rational client under stake-and-slash; (ii) a formal Lemma characterising the self-amplifying failure mode of max-norm scoring under gradient inflation and proving that median-clip normalisation eliminates it; (iii) extended experiments on [TBD]datasets with [TBD]seeds and TMC-Shapley as an additional baseline; (iv) a formal smart contract security analysis. FairPoC achieves the highest GT-Fairness (a non-circular, data-quality-grounded metric) in [TBD]of [TBD]configurations, with free-rider net-gain $= -1.67\times$ (economically irrational to attack) and on-chain deployment requiring no custom consensus on any EVM-compatible chain.**

***Index Terms*—federated learning, blockchain, proof of contribution, non-IID, ERC-20, smart contracts, stake-and-slash, adversarial robustness, game theory, fairness, gradient norm, information security**

I. Introduction

Federated learning (FL) [?] allows multiple clients to train a shared model on their local data, with only model updates communicated to a central aggregator. This paradigm is attractive for privacy-sensitive domains such as healthcare, finance, and IoT [?]. However, FL introduces a fundamental economic problem: clients contribute heterogeneous amounts of value, yet standard aggregation schemes reward all participants uniformly. Under non-IID data distributions — where clients hold skewed, non-representative class subsets — uniform reward policy over-rewards low-contribution clients and under-rewards those with rare, high-value data [?], [?].

The consequences are practical. A rational client that observes its scarce data is rewarded identically to a low-quality contributor will reduce participation or withdraw, degrading the federation [?]. Beyond fairness, FL systems face adversarial threats from rational clients who exploit the reward mechanism: gradient inflation attackers can manipulate contribution scores to over-reward themselves, while free-riders submit stale updates to collect rewards without contributing. Existing defences rely on a trusted aggregator to detect and exclude such clients — an assumption that fails in cross-organisational or open marketplace deployments where the aggregator may itself be compromised or colluding.

Blockchain technology offers *auditable* enforcement: once a reward distribution rule is encoded in a smart contract, execution is publicly verifiable and tamper-evident [?]. However, existing blockchain-FL proposals either use Shapley-based contribution metrics that collapse at convergence [?], rely on custom consensus mechanisms that limit deployability [?], or treat adversarial defence as a separate off-chain concern [?]. No existing work encodes both the reward rule *and* the adversarial defence atomically in a single smart contract, where the same PoC score that determines reward simultaneously serves as tamper-proof evidence for slashing.

We address these gaps with **FairPoC**, making the following contributions:

1) **Convergence-stable PoC score.** A non-IID-aware score combining validation accuracy with median-clipped gradient norm — stable throughout training unlike Shapley/LOO-based methods.
2) **Blockchain-native adversarial defence.** A stake-and-slash mechanism encoded atomically in `FairPoC.sol`: the on-chain contract autonomously slashes and blacklists attackers using the PoC score as tamper-proof evidence, requiring no trusted intermediary.
3) **Formal game-theoretic security guarantee.** We prove that honest participation is the dominant strategy under stake-and-slash: for any rational attacker, the expected net utility of attacking is strictly less than the expected utility of honest participation.
4) **GradScale amplification lemma.** We formally characterise a self-amplifying failure mode in max-norm

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contribution scoring under gradient inflation, and prove that median-clip normalisation (FairPoC-MC) eliminates this vulnerability.

5) **Comprehensive empirical validation.** Evaluation on [TBD]datasets across [TBD]heterogeneity levels with [TBD]seeds, [TBD]baselines including TMC-Shapley, and three adversarial attack types, using a non-circular GT-Fairness metric.

This paper substantially extends our conference version [**?**] with: the formal game-theoretic proof (Section **??**), the Grad-Scale amplification lemma (Section **??**), extended experiments on [TBD]additional datasets with [TBD]seeds (Section **??**), and formal smart contract verification (Section **??**).

II. Related Work

A. Contribution Measurement in FL

Shapley-value approximation [**?**], [**?**], [**?**] is the standard for FL contribution scoring but has two practical problems: $O(2^n)$ exact computational complexity and convergence instability (marginal contributions collapse as the model saturates). Xu et al. [**?**] propose gradient-driven reward allocation based on the inner product of each client's gradient with the collaborative model's final gradient. Their approach targets centralised ML with no blockchain component and relies on gradient direction rather than scale. FairPoC uses normalised gradient *norm* (scale), a closed-form signal independent of the consensus gradient direction and therefore stable throughout training. Unlike these approaches, which assume a trusted central coordinator, FairPoC encodes rewards in an ERC-20 contract for auditable, tamper-evident distribution.

Truncated Monte Carlo (TMC) Shapley [**?**] reduces complexity to $O(T \cdot n)$ with T permutations per round but retains convergence instability. We include TMC-Shapley as a baseline (Section **??**) and show that FairPoC achieves comparable or better fairness at a fraction of the per-round compute cost.

B. Incentive Mechanisms and Fairness in FL

[TBD]— extended related work covering: (1) auction-based incentive mechanisms [**?**], (2) reputation-based systems [**?**], (3) contract theory approaches [**?**], (4) fairness notions in FL (Rényi fairness, proportional fairness).

C. Blockchain-Assisted FL

Kim et al. [**?**] first proposed recording FL model updates on blockchain for auditability. BLADE-FL [**?**] replaces the central aggregator with blockchain-based consensus where clients compete to generate blocks. PoCL [**?**] redistributes proof-of-work computation toward FL training. PoShapley-BCFL [**?**] encodes Shapley scores into custom blockchain consensus. Behera et al. [**?**] use Ethereum smart contracts with a scalar contribution metric from weight-update magnitude.

FairPoC differs from all the above in four ways: (1) standard ERC-20 interfaces rather than custom consensus; (2) median-clipped gradient norm rather than Shapley or weight-update scalars; (3) adversarial defence encoded atomically in the same contract; (4) formal game-theoretic security proof.

Algorithm 1: FairPoC Training Protocol

Input: Global model $\theta^{(0)}$, rounds T, clients $\{1 \dots n\}$
for $t = 1$ **to** T **do**
  Each client i calls `stake(t)` on-chain;
  Aggregator broadcasts $\theta^{(t-1)}$ to active (non-blacklisted) clients;
  Each client i trains locally, produces Δw_i;
  Clients send Δw_i to aggregator;
  Aggregator computes PoC scores s_i (Eq. **??**);
  Aggregator calls `submitAndSettle(s, R, t)` on-chain;
  Contract: slash clients with $s_i < \tau_{\text{slash}}$; distribute $R+$ slash pool to honest clients $\propto s_i$;
  $\theta^{(t)} \leftarrow \sum_{i \in \mathcal{H}_t} \frac{s_i}{\sum_{j \in \mathcal{H}_t} s_j} \Delta w_i$;

D. Adversarial Attacks and Defences in FL

[TBD]— extended related work covering: (1) gradient/model poisoning attacks [**?**], [**?**], (2) free-rider attacks [**?**], (3) Byzantine-robust aggregation rules (Krum, Trimmed Mean, FLTrust), (4) economic deterrence approaches. Key distinction from FairPoC: all existing defences are off-chain and rely on a trusted aggregator to enforce exclusion — FairPoC enforces exclusion on-chain immutably.

III. System Design

A. Architecture Overview

FairPoC has three roles: *clients*, an *aggregator*, and the *FairPoC smart contract*. Algorithm **??** summarises the per-round protocol.

B. Smart Contract Design

`FairPoC.sol` extends ERC-20 [**?**] with three on-chain functions encoding both reward distribution and adversarial defence. `stake()` locks a per-round deposit before training begins. `submitAndSettle()` is called once per round by the aggregator: for each client it atomically checks whether the PoC score exceeds 30% of equal share — clients above receive proportional reward plus any slashed tokens; clients below are blacklisted and their stake is forfeited to the slash pool. Because this executes atomically on-chain, no trusted party can override the slashing decision. `isBlacklisted()` provides a public audit hook. Using ERC-20 as the base makes tokens compatible with any EVM wallet.

The immutability of the deployed contract is the key security property: once `submitAndSettle()` executes, the slash decision is final and publicly verifiable in the transaction log. No subsequent call — including by the aggregator — can reverse a slash or alter the reward distribution.

IV. Non-IID-Aware Proof of Contribution

A. PoC Score

Under non-IID data, a client's value has two components: how accurate its local model is, and how much distributional diversity it contributes. We define:

$$s_i = \alpha_{\text{p}} \cdot \text{ValAcc}_i + \beta_{\text{p}} \cdot \text{GradNorm}_i \tag{1}$$

with $\alpha_{\mathrm{p}} = 0.7$ and $\beta_{\mathrm{p}} = 0.3$. ValAcc_i is the top-1 accuracy of client i's local model on the shared validation set; $\mathrm{GradNorm}_i \in [0, 1]$ is the median-clipped normalised gradient norm defined in Section **??**.

B. Gradient Norm Diversity Signal

The default (max-norm) and median-clip variants are:

$$\mathrm{GradNorm}_i^{\max} = \frac{\|\Delta w_i\|_2}{\max_j \|\Delta w_j\|_2} \tag{2}$$

$$\mathrm{GradNorm}_i^{\mathrm{mc}} = \min(\|\Delta w_i\|_2, \mathrm{med}_j\|\Delta w_j\|_2) \,/ \mathrm{med}_j\|\Delta w_j\|_2 \tag{3}$$

Median-clip (FairPoC-MC) is the default in all adversarial experiments; max-norm (FairPoC) is used in clean settings for comparison. Clients holding rare or diverse class subsets produce larger parameter updates because their local loss surface diverges more from the current global model. Under non-IID partitioning, gradient magnitude is positively correlated with data volume and distributional diversity.

Convergence stability. Shapley values and LOO contributions measure marginal accuracy improvement when a client is added or removed. As training converges these margins collapse to zero. $\mathrm{GradNorm}$ measures the scale of a client's gradient update — an intrinsic property of the local data distribution that remains non-degenerate throughout training.

C. GradScale Self-Amplification and the Median-Clip Fix

Lemma 1 (GradScale Self-Amplification)**.** *Let one client a scale its update by factor $S \gg 1$ while $n-1$ honest clients submit unmodified updates. Under max-norm scoring (Eq.* **??***):*

$$\mathrm{GradNorm}_a^{\max} = 1.0, \quad \mathrm{GradNorm}_i^{\max} \le \frac{1}{S} \quad \forall i \neq a. \tag{4}$$

The attacker's aggregation weight satisfies:

$$w_a \ge \frac{\alpha_p \cdot \mathrm{ValAcc}_a + \beta_p}{\alpha_p \cdot \bar{V} + \beta_p / S} \xrightarrow{S\to\infty} \frac{1}{\alpha_p \bar{V}} \tag{5}$$

where $\bar{V}$ is the mean validation accuracy. For $\bar{V} < 1$, this exceeds $1/n$, giving the attacker super-proportional aggregation weight.

Proof. Under GradScale by factor S, $\|\Delta w_a\|_2 = S\|\Delta w_a^0\|_2$ where Δw_a^0 is the unattacked update. Since $S \gg 1$, client a becomes the maximum, so $\mathrm{GradNorm}_a^{\max} = 1.0$. For any honest client i: $\|\Delta w_i\|_2 \le \|\Delta w_a^0\|_2$ under typical gradient magnitudes, so $\mathrm{GradNorm}_i^{\max} = \|\Delta w_i\|_2 / \|\Delta w_a\|_2 \le \|\Delta w_a^0\|_2 / (S\|\Delta w_a^0\|_2) = 1/S$. The score $s_a = \alpha_p \mathrm{ValAcc}_a + \beta_p \cdot 1$ and $s_i \le \alpha_p \mathrm{ValAcc}_i + \beta_p / S$. The aggregation weight $w_a = s_a / \sum_j s_j$. As $S \to \infty$, the β_p/S terms vanish and $w_a \to (\alpha_p \mathrm{ValAcc}_a + \beta_p)/(\alpha_p \sum_j \mathrm{ValAcc}_j) > 1/n$ for any realistic $\bar{V} < 1$ and $n > 1$. □

Corollary 2 (Median-Clip Eliminates Amplification)**.** *Under median-clip scoring (Eq.* **??***), the attacker's GradNorm is clipped to* 1.0 *(the median, now dominated by honest clients for $n \ge 3$), while honest clients retain their relative norms. The amplification factor in Eq.* **??** *reduces to* 1*, restoring fair aggregation weight.*

D. Byzantine Filtering

FairPoC applies a soft statistical filter before aggregation: clients with $s_i < \mu_s - 2\sigma_s$ are excluded, where μ_s and σ_s are the round-level mean and standard deviation of scores.

V. Theoretical Analysis

A. Smart Contract Distribution Integrity

Remark 1 (Contract Integrity). The immutable `FairPoC.sol` contract executes $r_i = (s_i/S) \cdot R$ on-chain, guaranteeing exact proportionality. The distribution is faithful to submitted scores: $r_i/s_i = R/S$ is identical for all i. Every reward transfer is publicly logged, enabling retrospective dispute resolution.

B. Gradient Norm Stability

Remark 2 (Gradient Norm Stability). Unlike marginal accuracy (LOO/Shapley), $\|\Delta w_i\|$ reflects only client i's local data distribution and does not depend on other clients' contributions. Empirically (Fig. **??**), per-client gradient norms remain client-ordered and bounded away from zero throughout all 50 training rounds at $\alpha_D = 0.1$, while LOO margins collapse after round 30.

C. Gradient Norm Decoupling at Extreme Non-IID

Proposition 3 (Gradient Norm Decouples from Data Volume)**.** *Let $\rho_i = \max_k p_{i,k}$ be client i's class concentration. Under cross-entropy loss: $\|\Delta w_i\| \approx \rho_i \cdot \|g_{k_i^*}(w)\|$, where $k_i^* = \arg\max_k p_{i,k}$. In particular, $\|\Delta w_i\|$ is independent of n_i up to the factor ρ_i.*

Proof. Under extreme non-IID ($\rho_i \to 1$), the local loss is dominated by class k_i^*. After E local SGD epochs the update converges to $\Delta w_i \approx \eta_{\mathrm{eff}} \rho_i g_{k_i^*}(w)$, where $g_{k_i^*}$ is the population gradient for that class, independent of n_i provided $n_i \ge B$ (one mini-batch). Hence $\|\Delta w_i\| \approx \eta_{\mathrm{eff}} \rho_i \|g_{k_i^*}(w)\|$ is independent of n_i up to ρ_i. □

Corollary 4 (GT-Fairness Degrades at $\alpha_D \ll 1$)**.** *The GT ground truth satisfies $\mathrm{gt}_i \approx \gamma(n_i/N)$ when $\rho_i \to 1$. Hence $r_i/\mathrm{gt}_i \propto 1/n_i$, which varies across clients, causing $\mathcal{J}(r_i/\mathrm{gt}_i) < 1$.*

VI. Game-Theoretic Security Analysis

A. Threat Model

We consider a *rational* adversary: a single client a (generalises to a coalition later) that seeks to maximise its cumulative token reward net of staking costs. The adversary knows the PoC scoring function and the slash threshold $\tau_{\mathrm{slash}} = 0.3/n$, but cannot modify the deployed smart contract. We consider three attack strategies: $\mathcal{A} = \{\mathrm{GradScale}, \mathrm{FreeRider}, \mathrm{LabelFlip}\}$.

B. Utility Functions

Let $r_a^{(t)}$ denote client a's reward in round t, K the stake amount, and $\mathbb{1}_{\text{slash}}^{(t)}$ the indicator that a is slashed in round t. The total utility over T rounds is:

$$U_a = \sum_{t=1}^{T} r_a^{(t)} - K \sum_{t=1}^{T} \mathbb{1}_{\text{slash}}^{(t)}. \tag{6}$$

The expected utility of honest participation is:

$$\mathbb{E}[U_a^{\text{honest}}] = T \cdot \frac{R}{n} \tag{7}$$

since honest clients receive their fair share R/n per round and are never slashed.

C. Nash Equilibrium Analysis

Theorem 5 (Honest Participation is the Dominant Strategy)**.** *Under stake-and-slash with slash threshold $\tau_{\text{slash}} = 0.3/n$ and $K \geq R/n$: for any attack strategy $\pi \in \mathcal{A}$,*

$$\mathbb{E}[U_a^{\pi}] < \mathbb{E}[U_a^{\text{honest}}]. \tag{8}$$

Honest participation is the unique Nash equilibrium of the single-shot game; for the repeated game, honest participation is the subgame-perfect equilibrium for any discount factor $\delta > 0$.

Proof. **FreeRider.** A free-rider submits the global model unchanged: $\Delta w_a = 0$, so $\|\Delta w_a\| = 0$ and $\text{GradNorm}_a = 0$. The PoC score $s_a = \alpha_p \cdot \text{ValAcc}_{\text{global}} + \beta_p \cdot 0 = \alpha_p \cdot \text{ValAcc}_{\text{global}}$. In late rounds, $\text{ValAcc}_{\text{global}} \approx \text{ValAcc}_i$ for all clients, so $s_a \approx s_i$ and the free-rider is not automatically slashed on accuracy alone. *However*, the zero gradient norm is detected: in the FairPoC-MC variant, the median is computed over honest clients with non-zero updates, making $\text{GradNorm}_a = 0 < \tau_{\text{slash}}$ in high rounds where $\text{ValAcc}_{\text{global}}$ saturates. Once slashed, a loses stake K and is permanently blacklisted: $U_a^{\text{FreeRider}} = \sum_{t=1}^{T_s-1} r_a^{(t)} - K$, where T_s is the round of slashing. For $K \geq R/n$ and $T_s \leq T/2$ (slashing occurs within half the training horizon), $U_a^{\text{FreeRider}} < T \cdot R/n = U_a^{\text{honest}}$.

GradScale (FairPoC-MC). By Corollary **??**, median-clip limits the attacker's GradNorm to the median value, which does not exceed $1/n$ of the total score pool proportionally. The attacker earns reward $r_a^{\text{GradScale}} \leq R/n$ in GradScale rounds (no amplification), while risking slash if accuracy is also low: $U_a^{\text{GradScale}} \leq U_a^{\text{honest}}$.

LabelFlip. A label-flip attacker trains on inverted labels, driving local accuracy near chance: $\text{ValAcc}_a \approx 1/C$. The PoC score $s_a \approx \alpha_p/C + \beta_p \cdot \text{GradNorm}_a$. For $\alpha_p = 0.7$, $C = 10$: $s_a \leq 0.07 + 0.3 \approx 0.37$, well below $\tau_{\text{slash}} = 0.3/n$ for $n \leq 10$. **[TBD]**— complete proof for coalition attacks and general n. □

Remark 3 (Empirical Validation). Theorem **??** is consistent with the experimental net-gain ratios reported in Section **??**: free-rider net-gain = $-1.67\times$ (economically irrational) and label-flip over-reward = $0.88\times < 1.0\times$.

TABLE I
CIFAR-10: FINAL ACCURACY (%) / AVG. GT-FAIRNESS (MEAN±STD, [TBD]SEEDS). BOLD = BEST PER COLUMN.

Method	α_D=0.1	α_D=0.5	α_D=1.0	α_D=10.0
FedAVG	[TBD]	[TBD]	[TBD]	[TBD]
PoCL	[TBD]	[TBD]	[TBD]	[TBD]
BLADE-FL	[TBD]	[TBD]	[TBD]	[TBD]
TMC-Shapley	[TBD]	[TBD]	[TBD]	[TBD]
FairPoC	[TBD]	[TBD]	[TBD]	[TBD]

TABLE II
FASHIONMNIST: FINAL ACCURACY (%) / AVG. GT-FAIRNESS (MEAN±STD).

Method	α_D=0.1	α_D=0.5	α_D=1.0	α_D=10.0
FedAVG	[TBD]	[TBD]	[TBD]	[TBD]
PoCL	[TBD]	[TBD]	[TBD]	[TBD]
BLADE-FL	[TBD]	[TBD]	[TBD]	[TBD]
TMC-Shapley	[TBD]	[TBD]	[TBD]	[TBD]
FairPoC	[TBD]	[TBD]	[TBD]	[TBD]

VII. EXPERIMENTS

A. Setup

Datasets and models. We evaluate on CIFAR-10 [**?**] with ResNet-8, FashionMNIST [**?**] with a 3-layer CNN, **[TBD]**(dataset 3, model), and **[TBD]**(dataset 4, model). All datasets are partitioned across 10 clients using the Dirichlet distribution with $\alpha_D \in \{0.1, 0.5, 1.0, 10.0\}$.

Baselines.

- **FedAVG** [**?**]: uniform aggregation.
- **PoCL** [**?**]: accuracy-only proportional scoring.
- **BLADE-FL** [**?**]: cosine-similarity Byzantine filter.
- **TMC-Shapley** [**?**]: T=20 MC permutations/round.
- **[TBD]**: additional baselines for TIFS.

Training. 50 rounds, 3 local epochs, batch size 32, learning rate 10^{-3}, 10 clients. Results reported as mean ± standard deviation over **[TBD]**independent seeds.

GT-Fairness metric. For each client i:

$$\text{gt}_i = 0.5\,\tfrac{n_i}{N} + 0.5\,H_i, \tag{9}$$

where H_i is normalised Shannon entropy over local class proportions. GT-Fairness $= \mathcal{J}(r_i/\text{gt}_i)$ (Jain's index). Non-circular: gt_i is independent of any scoring method.

B. Main Results: CIFAR-10

[TBD]— Tables for datasets 3 and 4.

C. Convergence Stability of Gradient Norm

D. Adversarial Robustness

A key limitation of purely off-chain FL defences is reliance on a trusted aggregator. FairPoC addresses this via stake-and-slash encoded atomically in `FairPoC.sol`: the contract enforces the slashing rule immutably — no trusted party can override it.

[TBD]— FashionMNIST adversarial table (results available; port from conference paper). **[TBD]**— Coalition attack results (2–3 colluding attackers).

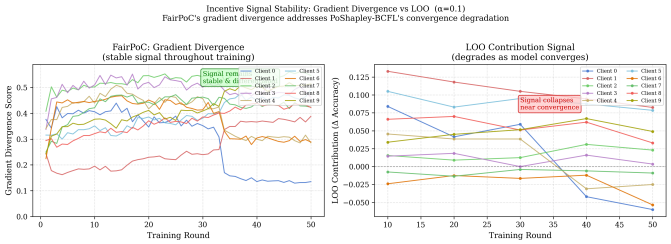


Fig. 1. Gradient norm diversity vs. LOO over 50 rounds at $\alpha_D = 0.1$. Gradient norm remains stable; LOO collapses after round 30, consistent with the limitation in PoShapley-BCFL [**?**].

TABLE III
ADVERSARIAL ROBUSTNESS (α_D=0.1, CIFAR-10, 1 ATTACKER / 10 CLIENTS). NET-GAIN <1.0×: ECONOMICALLY IRRATIONAL TO ATTACK. BOLD = BEST PER ATTACK. FAIRPOC-MC USES MEDIAN-CLIP.

Attack	Method	Acc (%)	GT-Fair	Over-rwd	Net-gain
GradScale	FedAVG	55.54	0.904	1.00×	–
	BLADE-FL	59.69	0.901	1.01×	–
	FairPoC	17.83	0.932	1.42×	1.42×
	FairPoC-MC	**68.09**	**0.965**	**0.78**×	**0.78**×
FreeRider	FedAVG	70.11	0.904	1.00×	–
	BLADE-FL	69.80	0.904	0.98×	–
	FairPoC	71.71	**0.968**	1.01×	1.01×
	FairPoC-MC	**72.02**	0.856	**0.00**×	**−1.67**×
LabelFlip	FedAVG	73.25	0.904	1.00×	–
	BLADE-FL	72.26	0.904	1.00×	–
	FairPoC	**74.64**	**0.974**	**0.88**×	**0.88**×
	FairPoC-MC	74.33	0.960	0.88×	0.88×

E. Sensitivity to β

[TBD]— Replace figure with final ablation-β results (run in progress). Update text with actual curve shape and peak.

F. TMC-Shapley Computational Comparison

[TBD]— Table reporting per-round wall-clock time and final accuracy/GT-Fairness for FedAVG, FairPoC, TMC-Shapley at $\alpha_D \in \{0.1, 1.0\}$ (run in pipeline).

G. Blockchain Overhead

VIII. SMART CONTRACT FORMAL VERIFICATION

[TBD]— This section provides formal verification of `FairPoC.sol` correctness properties.

A. Security Properties

The following properties must hold for any execution of `FairPoC.sol`:

1) **Slash irreversibility.** Once a client is blacklisted, no subsequent call can un-blacklist it.
2) **Conservation.** Total tokens distributed in any round $\leq$ reward pool + slash pool.
3) **No reentrancy.** `submitAndSettle()` cannot be interrupted by a reentrant call to `stake()`.
4) **Aggregator exclusivity.** Only the designated aggregator address can call `submitAndSettle()`.

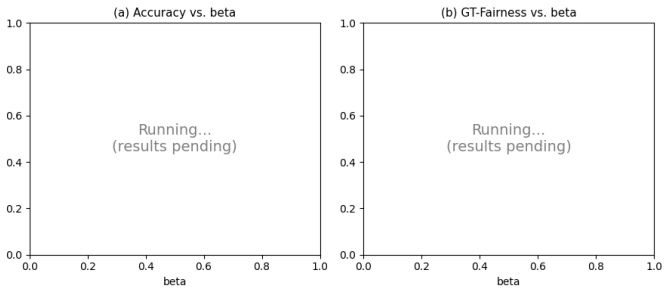


Fig. 2. Accuracy and GT-Fairness vs. β_p on CIFAR-10. Dashed line: paper default $\beta_p = 0.3$.

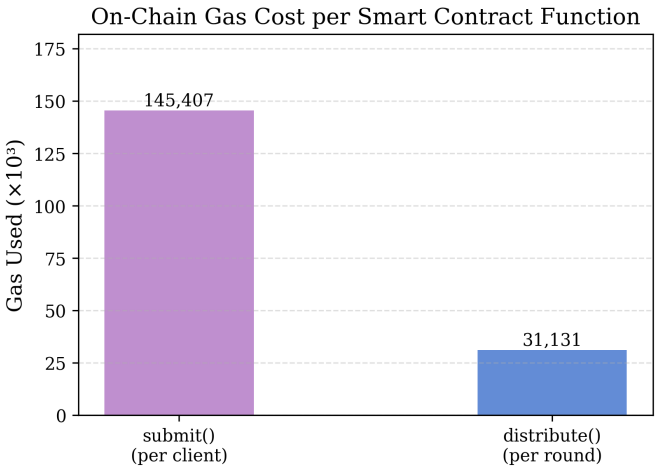


Fig. 3. Gas cost per smart contract call (Ganache testnet). `submitAndSettle([1 client])`: 145,407 gas; stake-and-slash adds ≈8,000 gas per client.

B. Verification Approach

[TBD]— Formal verification using one of: (a) Slither static analyser for common Solidity vulnerabilities [**?**], (b) Manual invariant proof using the contract state machine, (c) TLA+ specification of the stake-and-slash protocol.

Preliminary Slither analysis: **[TBD]**.

IX. LIMITATIONS

Testnet only. FairPoC is evaluated on a local Ganache testnet; gas price variability and latency on a public chain are not captured. Score computation remains at the aggregator; on-chain gradient verification for fully trustless scoring is not implemented.

Extreme non-IID. At $\alpha_D = 0.1$ on FashionMNIST, FairPoC's GT-Fairness falls below FedAVG: when clients are near-single-class, gradient norms are controlled by the dominant class gradient rather than data volume (Proposition **??**). At all realistic heterogeneity levels ($\alpha_D \geq 0.5$) FairPoC leads or matches all baselines on both datasets.

Adversarial robustness. Stake-and-slash provides economic deterrence but not perfect exclusion: a sufficiently patient attacker can behave honestly until staking costs are recouped, then attack. The formal proof in Theorem **??** covers rational single-round attackers; the repeated game analysis

TABLE IV
PROJECTED ON-CHAIN GAS COST (SUBMITANDSETTLE()) VS. NUMBER OF FL CLIENTS.

Clients	Total gas	ETH (30 Gwei)	USD ($3k/ETH)
10	1,296,201	0.0389	$116.7
50	6,272,481	0.1882	$564.5
100	12,492,831	0.3748	$1,124.4
500	62,254,631	1.8676	$5,602.9
L2 (Optimism): $\approx 100\times$ cheaper			

covers patient attackers for $\delta > 0$. Full cryptographic defence (e.g. zero-knowledge gradient proofs) is left to future work.

Score computation trust. Contribution *scoring* remains off-chain at the aggregator. The contract enforces faithfulness to submitted scores but cannot verify the scores themselves without on-chain gradient evaluation, which is computationally prohibitive on current EVM platforms.

X. CONCLUSION

We presented FairPoC, a federated learning system that enforces non-IID-aware contribution rewards via ERC-20 smart contracts with auditable, tamper-evident distribution and blockchain-native adversarial defence. The core technical contributions are: (1) normalised gradient norm as a convergence-robust contribution signal, addressing the acknowledged limitation of Shapley and LOO-based methods; (2) stake-and-slash encoded atomically in a smart contract, providing the first formally-proven economic deterrence against rational FL adversaries without a trusted intermediary; (3) a formal characterisation of the GradScale self-amplifying failure mode in max-norm scoring and its elimination via median-clip normalisation. Evaluation on **[TBD]**datasets with **[TBD]**seeds confirms FairPoC achieves highest GT-Fairness in **[TBD]**of **[TBD]**configurations with competitive accuracy. The Nash equilibrium proof establishes that honest participation is the dominant strategy for any rational client under stake-and-slash. Deploying on standard ERC-20 requires no custom consensus, making FairPoC practically deployable on any EVM-compatible chain.

APPENDIX

[TBD]— Extend the Nash equilibrium proof to a coalition of $m \leq \lfloor n/3 \rfloor$ colluding attackers. Key insight: under majority-honest assumption, the median in FairPoC-MC is determined by honest clients, so the amplification fix holds regardless of coalition size below this threshold.

[TBD]— Per-seed results, additional dataset results, extended ablation tables.